

Millisecond-latency Visual Fault-buttons using Event-cameras

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Abstract. Workplace safety, during interaction with automated machinery and robots, can be improved by monitoring both robot and human actions to reduce accidents and injury. Crucial to safety is a low latency detection system. Here we propose to decrease latency by detecting the intention of an operator to press a fault button of an industrial robot, so that it is stopped before the fault button is physically pressed. To do so, we exploit an extremely low latency, vision sensor - i.e. event cameras - to detect fast motion in the fault-button area. We prove that our system can initiate a shutdown up to 500 ms before the physical button is pressed. We analyse the parameters of the algorithm with respect to three different industrial scenarios and environments, and propose an adaptive threshold algorithm to reduce tuning requirements. An interactive application is provided to easily deploy the proposed visual fault-button detection.¹

Keywords: event camera · low latency · monitoring · workplace safety

1 Introduction

Fault buttons are used to immediately halt potentially dangerous machinery in emergency situations to prevent damage to operators and the machines themselves. Hard fault buttons are electrically connected to the power of the machine and physically stop electricity from reaching actuators. Soft fault buttons stop machine actuation from within the machine’s software controller. Typically a mix of hard and soft fault buttons may be used in an automated setting.

The difference between emergency situation and near-miss can be a matter of milliseconds, but an operator may not always be immediately beside a fault button. Early prediction of fault button intention with visual sensors can reduce the time taken between when an emergency is perceived by the operator, and when a fault button is triggered.

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Even an operator standing within arms reach of a fault button can take over 500 ms from initial arm motion, until when the button is pressed. Event-cameras have millisecond latency and their design makes them particularly adept at processing high-speed visual stimuli. We propose a visual fault button using event-cameras to preempt the operator activating a wired fault button and activate the soft fault button.

Our method capitalises on the technology of event-cameras such that, in combination with the specific application of the early detection of a fault button press, only simple algorithm is required, which also reduces the latency of the data processing.

This paper contributes:

- The first investigation into using event-cameras for workplace safety, in particular for early fault-button detection.
- Methodology for low-latency detection of fault-buttons using event-rate, together with an analysis of parameters and adaptive thresholds.
- Datasets, including three real-world industrial scenarios, and code made available open-source.



Fig. 1: Operator activating an hardwired fault button as observed by an event camera: (a) each dot represents a digital event in space and time (purple ON, green OFF events). (b) events in a small time window accumulated on a single frame, the polarity is not represented. The red circle shows the region of interest around the fault button used by for the algorithm.

2 Related Work

Video cameras are currently used in industrial sites for workplace safety to identify potential hazards, identify and monitory equipment failure and damage, and aid in investigations after incidents occur [10]. Collision avoidance systems can

rely on depth sensors to compute the distance between operators and robots [6], other methods propose more on adaptive systems, such as virtual repulsive fields [11] or on direct control of the robot for collision avoidance [13, 15, 16]. Motion forecasting has also been used to modify robot action to improve safety [1]. Lighting variation and system latency have been identified as specific challenges with such visual systems [10].

All these systems attempt to prevent workplace incidents as a very important step towards workplace safety. Additionally, reducing impact when inevitable contact does occur has been investigated [2–4]. Improving emergency systems after an incident has occurred augments such safety systems, but has received less investigation. The correct visual technology may not have been available to do so.

In all these scenarios, and also in post-incident emergency response, high-frequency and low-latency improve the outcome of the safety system. Event-cameras [12] have independent circuitry per pixel, allowing each pixel to monitor for a change in light, and signal precisely when it occurs. The camera produces a signal only when a pixel detects a large enough change in brightness allowing a small, low-latency, packet of data to be read by processing algorithms; instead of sending the absolute brightness for every pixel, every frame, as with a traditional camera. They also do not have a global exposure time, thereby also achieving a higher dynamic range than traditional sensors for both bright and dark (or mixed) environments.

While event-cameras haven’t been investigated for work-place safety applications, instead intruder monitoring has relied on the ability to segment moving targets based on the strong visual change signal [14]. The low-latency and lite-weight processing has been explored for robotic control [8, 9] including drones [5, 17]. Event-cameras have shown to have at-least an order of magnitude lower latency [7] than traditional cameras.

Therefore we propose the use of event-cameras for workplace safety, in particular investigating low-latency detection of an incident through human intention to press a fault-button.

3 Methods

The visual fault button (VFB) has multiple components:

- A stationary event camera mounted to keep an operator and a wired fault button in view, in non-overlapping image space.
- A defined fault-button region ($R = \langle u, v, r \rangle$) in visual space, that can be calibrated at start-up; where $\langle u, v \rangle$ is the pixel position of the wired fault-button and r is a radius in pixels, see Fig. 1.

The VFB monitors signal frequency within R . Actions to press fault buttons are typically faster, and therefore generate a higher number of events in a given spatio-temporal window, than other standard operating actions. Event-cameras

can monitor the frequency of the signal and detect fault-button pressing intent in milliseconds.

The state of the VFB, x_{vbf} is activated if:

$$x_{vbf} = \begin{cases} 1, & \text{if } \sum v_{t-k}^t > T \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

where v is an event within R , t is the current time, k is an algorithm parameter defining a temporal window, and T is a calibrated threshold. k is a fixed parameter defined through experimental procedure, while R and T are defined during a calibration procedure. T can be further dynamically calibrated during run-time.

When the VBF is activated, i.e. $x_{vbf} = 1$, the safety systems of the operating machines are engaged.

3.1 Parameter Selection Methods

Methods can be used to determine reasonable parameters for the VFB.

Temporal Bin Size (k) is dependent on the relationship between temporal dynamics in the scene and the temporal dynamics of the sensor. The temporal dynamics of the scene can be assumed to be relatively constant (i.e. the typical speed of human movements), however the temporal dynamics of the sensor vary between sensors as well as due to lighting conditions. The parameter is best set after analysis of example data for each installed VFB set-up.

Region Size (r) is dependent on the placement of the camera relative to the fault button and the expected amount of motion in the captured region, e.g. from other workers and machinery/robot actuation. The region position and size should be set manually on a case-by-case basis for each VFB. A simple procedure of clicking and dragging to define the region is implemented in the provided application. A bigger region's size can lead to earlier detections but at the cost of decreased robustness.

Threshold (T) is dependent on many factors including lighting, noise, relative size of moving objects in the camera, speed of the worker triggering the fault button, and speed of workers/machinery operating within the region R . The threshold parameter can be adaptively tuned on-the-fly during operation to remove the need to hand-tune the parameter and improve detection robustness and accuracy.

3.2 Adaptive Threshold

The threshold T must be tuned based on analysis of the event-rates for non-button presses, compared with event-rates for actual button presses. Therefore

it is possible to propose an adaptive threshold that is intelligently set from data measured during operation.

To calculate an adaptive threshold, two sets of event-rates are first generated:

- a set of trigger rates (RT); containing all rates measured in a period before a measure fault button is pressed.
- a set of inactivity rates (RI); containing all rates measured during normal operation. The distribution of this set has a high variance as it can be zero (no motion) with many spikes in event-rate as inconsequential activity occurs within the R .

Event-rate samples are added to each distribution at a small (1 second) delay. An event-rate is added to

- trigger rates: if the button has been pressed in the last second. The top 50% of rates in the last second are added and the buffer cleared; and
- inactivity rates: if no button has been pressed and the rate is higher than the current threshold level. This effectively stores only the higher inactivity rates which are more important for determining the threshold, the lower rates are not important. It also allows for lower memory utilization and faster execution.

To completely avoid false detection, the threshold value could be selected to be the maximum event-rate not associated with a fault-button press. However, any large event-rate without a button press renders the system in-able to further preempt true button presses.

Instead, the problem can be framed as a small optimization with the goal to minimize both false positives f_p and false negatives f_n . Specifically, for a given threshold T :

$$f_n = \sum_{i=0}^{RT_z} RT_i < T \quad (2)$$

where RT_z is the size of the set RT , and

$$f_p = \sum_{i=0}^{RI_z} RI_i > T \quad (3)$$

where RI_z is the size of the set RI .

The loss function is set as:

$$loss = \frac{f_p}{RI_z} w + \frac{f_n}{RT_z} (1 - w) \quad (4)$$

The number of missed and false detections is normalized by the number of trigger rates and inactivity rates respectively. Each component has a weight that depends on w . A higher value of w will prioritize false detections. For our tests, we use a value of $w = 0.8$.

3.3 Application

A software application that allows the user to select the fault button location, and define the active-region of the fault-button easily is available for testing².

4 Datasets

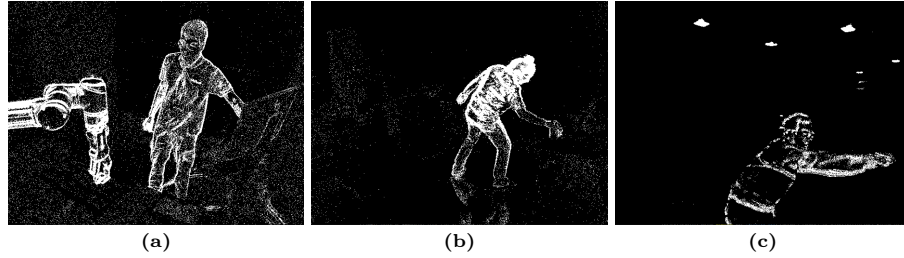


Fig. 2: Examples of each of the industrial scene datasets labelled (a) Dataset 1, (b) Dataset 2, and (c) Dataset 3. Details are given in Section 4

Early detection of visual fault buttons is a novel concept and to the author’s knowledge no other datasets exist, especially not with event cameras. To evaluate the effectiveness and benefit of a visual fault button, the method was evaluated on several different industrial conditions, involving machinery, an operator and a wired fault button. The datasets were taken in real industrial settings in real applications with robots on production lines. Datasets have been made available to the community³.

- Dataset 1 consists of 4 sequences of the event camera observing a worker standing in the approximate centre of the image. The left side of the scene includes a moving robotic arm, while the fault button is positioned on the right of the operator. No other motion occurs near the fault button, except for the operator pressing the fault button. The sequences involve 400 seconds of data with 12 fault presses. An example is shown in Fig. 2a.
- Dataset 2 does not contain any other moving objects, only the operator moves. The operator also performs other tasks in various part of the visual field, even behind the fault button. The sequences add up to 175 seconds with 12 fault presses. An example is shown in Fig. 2b.
- Dataset 3 presents a more challenging scenario. The button is placed on top of a conveyor belt transporting objects. This can cause relatively fast motions close to the button that are not directly caused by the operator. Furthermore, the dataset presents more challenging illumination and the

² <https://github.com/event-driven-robotics/visual-fault-button>

³ <https://zenodo.org/records/13358681>

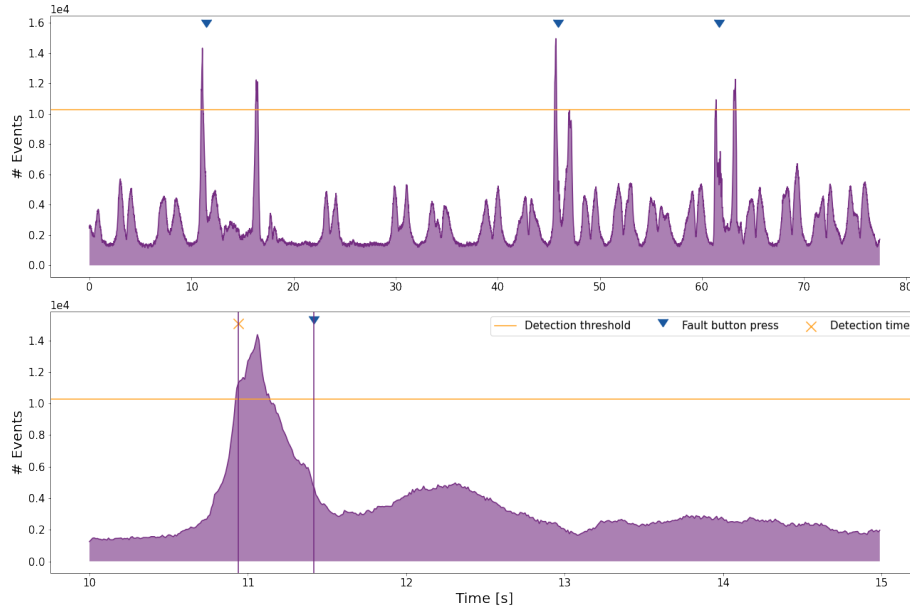


Fig. 3: Event rate over time within R for the dataset 1, with parameters $R = 184$ pixels, $T = 11.7$ e/s, and $k = 0.01$ seconds. The small delay between the event rate crossing the threshold and the detection time is caused by the size of the detection bins. Rate measurements are available only at specific intervals which can cause some delays in the detection.

camera focus is not optimal. The total duration is 90 seconds with 8 fault presses. An example is shown in Fig. 2c.

The datasets do not contain groundtruth information about the fault button presses. Instead, the sequences were manually labelled with the time of each fault button press. The high temporal resolution of event-cameras allows higher precision labelling compared to frame-based traditional cameras.

5 Experiments

No other methods for visual fault button detection exist and instead, to evaluate the proposed algorithm, an ablation study was performed. We analyse the performance of the algorithm depending on the parameters k , T , and r ; with $\langle u, v \rangle$ centered at the fault button pixel location. False positive and false negative detections provide a metric for system accuracy. The datasets are analysed to find the average temporal safety margin (TSF): the time between visually detected intent and the final triggering of the fault button by the human operator.

5.1 Temporal Bin Size

For each dataset, we manually set the region size r and the detection threshold T . We vary the temporal period k and observe the detection accuracy and the TSM. A k of approximately 1 ms gives an optimal value to reduce false positives and maintain as large a TSM as possible.

For all the datasets the TSM becomes shorter for larger temporal bin sizes, as shown in Figure 4a, but only decreasing by $\approx 100\text{ms}$ from 0.1 ms to 100 ms bin sizes. The larger bin size acts as a low-pass filter and it takes more data to reach the peak value. The variation in TSM between datasets occurs as we employ different region sizes r for each dataset due to variation in the scenarios.

Smaller bin sizes increases the high-frequency noise of the signal. For all datasets, false detections increases for bins less than 1 ms. The consistency of this value across datasets indicates a dependency on the sensor itself, which tends to have a temporal jitter (i.e. minimal accuracy) on the order of 1 ms.

5.2 Region Size

k and T were held constant (manually selected for each dataset) and the region-size r was varied. The value of r should be chosen based on individual conditions for each VFB set-up, and should typically be chosen to be as large as possible while rejecting false positive detections.

The size of the detection region influences how far away from the fault button intention can be detected, and therefore translates into how early detection can occur. A larger region allows the detection of fast movements earlier, but it may introduce more possibility of false detection, as movements further away from

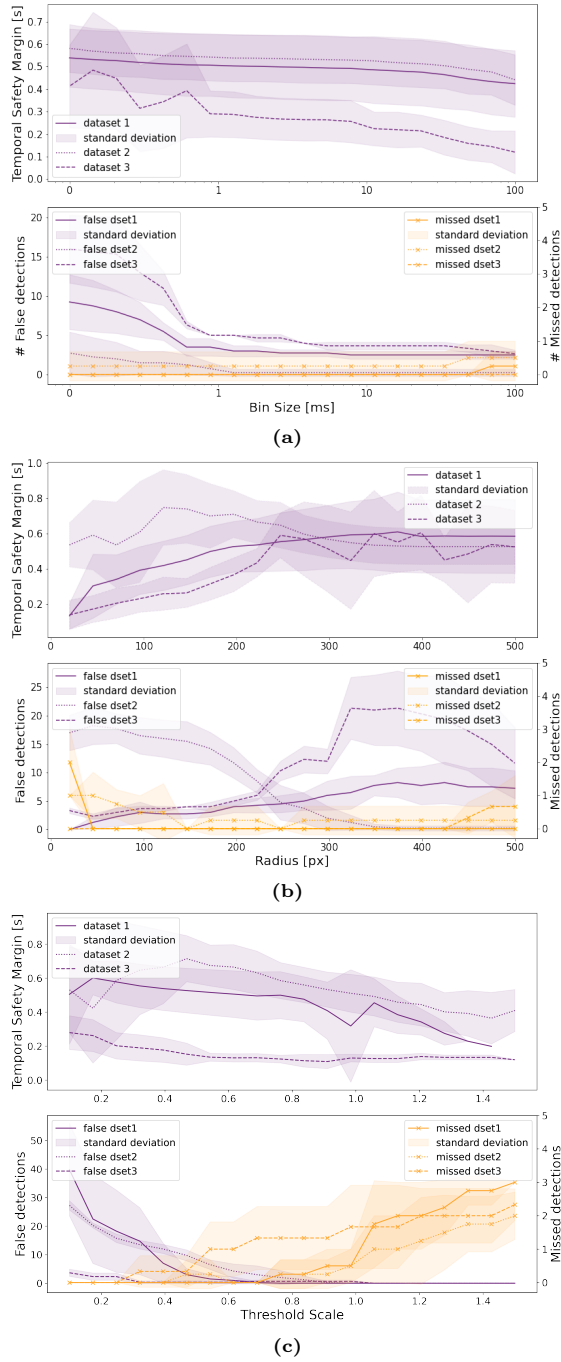


Fig. 4: Temporal Safety Margin (above) and accuracy (below) as the (a) temporal bin size, k , (b) region size, r , and (c) threshold, T , are modified. Figures show the mean and variance split per dataset.

the button are less likely to be caused by the operator moving to press. As such a clear increasing trend of TSM with r can be seen in Fig. 4b.

False positive detections can also increase with a larger r depending on the amount and type of background activity. In general a selection of r between 200 and 300 pixels should maximise TSM while minimising false detections.

However Dataset 2 shows a divergent trend compared to Dataset 1 and Dataset 3. Dataset 2 has both activities performed with the hands nearby the fault button, combined with often large bodily movements required to hit the fault button. A small value of r requires fault-button detection based on hand-movement alone, which is somewhat difficult due to the pre-described activities. A large value of r , instead, can use cues from whole body motion to detect fault-button intention.

5.3 Threshold

The values of k and r were held constant (manually selected for each dataset) and the threshold T was varied. The threshold T is dependent on the environment and set-up of the VFB but also highly dependent on lighting conditions and un-measurable properties of the sensor itself and therefore requires tuning. Thresholds vary between datasets, and the x-axis of Fig. 4c is instead shown with a relative threshold value, in which the value of 1.0 is set to the event rate of the first button press of each dataset.

While the TSM decreases with a higher threshold, as it takes (slightly) more measurements to reach a higher threshold value, T , must be selected to reject all non fault button motions that occur in the R , while not rejecting true button presses. All datasets show a similar trend in that a low threshold produces false positive detections, while a high threshold results in missed detections. The ideal threshold both minimizes false and missed detections and an adaptive threshold is possible to achieve this.

5.4 Adaptive Threshold

We tested the adaptive threshold method on all 3 datasets. We merged all the sequences of a dataset into a single one and observe the number of missed and false detections. The method needs some time to find a good threshold value. To test the quality of the selected value we run the method once on the whole sequence, to find a good threshold. Then we repeat the sequence again and measure the detection statistics. Table 1 shows the results for each of the datasets.

6 Discussion

The proposed algorithm is based on simple features of the event-camera data stream (i.e. event-rate). Therefore, the robustness of the system assumes that large, high-velocity entities do not pass through R except under fault-button-reaching conditions. If the assumption is not valid, e.g. many operators running

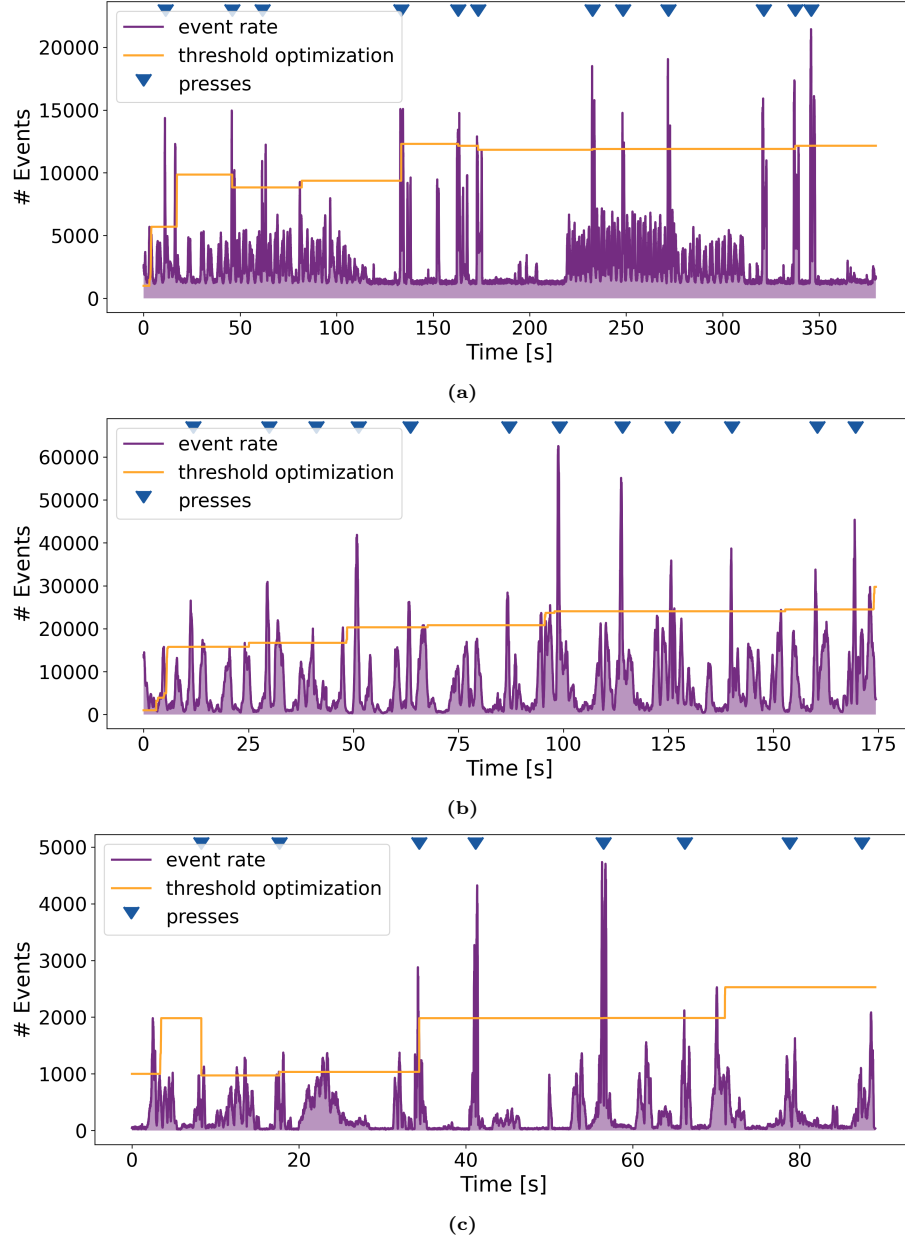


Fig. 5: Adaptive Threshold shown over time for (a) Dataset 1, (b) Dataset 2, and (c) Dataset 3.

quickly through the region R , the proposed visual fault button may not be suitable.

Robustness could be improved by using higher-level features or more complex methods, for example:

- Velocity could be extracted from the events passing through R to identify entities that move quickly, and in the direction toward the fault button.
- Human pose estimation could be used to identify hands and arms and only detection the motion of these limbs when considering fault button intention.
- Deep neural networks could be trained to extract signals related to fault button intention not obvious from observation.

However, more complex algorithms increase reducing the resulting TSM. For instance the open-source human pose detection network, OpenPose, can have over 100 ms latency, thereby reducing the TSM by 20%. Such algorithms can also require large computational clusters with GPU computation to realise, while the proposed system is simple enough to run on small embedded hardware.

The system currently assumes the region R is circular centred on the fault button, however specific workplace set-ups may require custom shapes and sizes, for example if machinery operates close to one side of the fault button, but not on the other. A custom image mask that defines R may instead demonstrate better performance under such conditions.

The final parameter selection, especially the Threshold T , should take into consideration human nature in operator behaviour. A system that is incorrectly triggering safety protocols too often may be turned off entirely as it interrupts work-flow. Discussion with workers and monitoring of such a system may be required to deploy a system actually beneficial to a workplace. False detections also increase the operational cost, as parts in production might have to be scrapped and a large amount of time would be necessary to restore the system to normal operation.

7 Conclusion

A visual fault button (VFB) algorithm was presented in which an event-camera can augment the physical fault button required for any machinery and industrial automation. The VFB showed to increase the temporal safety margin (TSM), i.e. how early the intention of pressing the fault button can be detected, by an average of 0.452 s. Algorithm parameters: the temporal bin size k should be set based on the temporal sensor accuracy (1 ms), while the region size should be set on a case-by-case basis depending on the industrial environment and camera view. An adaptive threshold was proposed to optimise the false-positive to missed-detections on-the-fly.

The event-camera is a highly suited visual sensor for VFB as it offers extremely low latency, allowing visual detection *before* the fault button is physically pressed, and high-temporal resolution, allowing relatively robust feature extraction based solely on temporal statistics.

Dataset	1	2	3
Length (s)	379	174	89
Presses	12	12	8
False Detections	0	0	0
Missed Detections	1	4	5
Mean TSM (s)	0.482	0.509	0.196

Table 1: Detection and temporal safety margin achieved using adaptive threshold and optimised parameters.

Low-computation and low-power solutions can be realised using event-cameras, to increase safety in industrial situations, to reduce severity of injury in the workplace. This pilot study shows the untapped potential for such systems, while live trials in real factories are required to fully understand the benefit.

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